Project Walkthrough for Bu Dinglin

A plain-English guide to the RAM/SSD and AI bubble project

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The big idea

One sentence: AI data centres are buying up the world's memory, so RAM and SSD prices have roughly quadrupled. Whether that continues depends on whether AI spending is a lasting boom or a bubble.

This is our Data Visualisation course project. We submit two things: the **video** (6 minutes max) and the **2-page Summary Report**. Everything else supports those:

- `summary.pdf`: the 2-page Summary Report we submit. Individual contributions are listed in Section 5.
- `RAM_AI_Bubble.pptx`: the deck we present and record from. Each slide plays one animated chart, and the narration is in the speaker notes.
- `video_script.pdf`: the same narration as a printable script. Ayush does the intro and the data story (Scenes 1–5); you take over at Scene 6 and do the conclusion.
- `clips/`: the eleven animated charts as MP4 files, in case we edit the video outside PowerPoint.
- `report.pdf`: the full argument with every source (9 pages). Use it to check any number.
- `data/`: the three datasets we built, as CSV files.

The data we built

The course asks us to explore a dataset, so we compiled three time series from public sources:

- **DDR4 chip contract price**, monthly, Jan 2022 to Sep 2026 (DRAMeXchange via Silicon Analysts). This is the star chart: it shows the 2022–23 bust ($3.41 down to $1.30) and the AI boom (up to $26, 20× the low).
- **Big Tech capex**, quarterly, 2022–2025, for Microsoft, Amazon, Alphabet, Meta and Oracle (Epoch AI, from SEC filings). It rose from $37.6B to $140.6B a quarter (3.7×). This is the demand behind the shortage.
- **SK hynix operating profit**, quarterly, 4Q22 to 2Q26 (company releases): from a 3.4tn-won loss to a 60.5tn-won profit. This shows who wins.

One caveat to know: DDR4 is being phased out by manufacturers, which exaggerates its price spike. Newer memory (DDR5) rose too, just less dramatically. The script says this, so keep it in.

Glossary: the words you'll hear

Term	Meaning
DRAM	Working memory. The “RAM” in a PC, phone or server.
NAND flash	Storage chips, the inside of an SSD, phone storage or a memory card.
HBM	High-bandwidth memory. A stacked, very fast DRAM used in AI chips. Highly profitable, and it uses up factory capacity.
Contract price	What big buyers like PC makers pay under supply deals. More reliable than shop prices.
Spot price	The price for one-off, immediate purchases. Moves faster, and often turns first.
Hyperscaler	The giant cloud companies: Amazon, Microsoft, Alphabet (Google), Meta and others.
Capex	Capital expenditure: money spent on long-lasting things like data centres and chips.
Depreciation	Spreading the cost of equipment over its useful life. AI chips are written off over about six years.
Bill of materials	The total cost of the parts in a product.

The story, section by section

1. How the memory market works

Only three companies, Samsung, SK hynix and Micron, make about 88% of the world’s DRAM. So when they change what they make, prices move for everyone. Memory also swings between shortage and glut: in late 2022, prices roughly halved and SK hynix made a record loss.

2. Why prices rose

It’s a four-step chain. AI spending exploded; memory makers shifted factories to the more profitable AI memory (HBM and server DRAM, now about half of DRAM supply); cloud giants locked in supply with long-term deals; and SSDs followed. The demand comes from data centres, so weaker laptop sales can’t fix it.

3. How big the increase is

A 32GB DDR5 kit went from about $90 to $425 in a year. Contract DRAM prices rose 58–63% in one quarter, and NAND 70–75%. Some prices are starting to soften: NAND spot prices dipped about 1% in September 2026.

4. Who pays

Mostly consumers. Gartner expects PC prices to rise 17% in 2026. Memory can be close to 60% of a $400 phone’s parts cost. Smaller manufacturers have the least buying power.

5. When it eases

Not before 2027 for RAM. New factories add about 27% more server memory in 2027, but SK hynix warns 2027 could be the tightest year yet. A Chinese competitor, CXMT, is ramping up.

6–7. The AI bubble debate

Big Tech capex doubles to about $1 trillion in 2026. The worry: that spending equals the cloud revenue meant to pay for it, depreciation will balloon, and debt is rising. The defence: valuations are far below the 2000 dot-com peak, demand is real, and most of it is paid from profits. Our middle view is that AI is real but probably overbuilt for a few years: write-downs, not a crash.

8. How the two connect

Memory prices are a bet on AI spending. Boom continues: RAM stays expensive. Slowdown: prices drift down. Bubble deflates: prices could crash, as in 2022. These scenarios are our reasoning, not a sourced forecast, so always say “if”.

Questions we might get, and how to answer

“Why don’t memory makers just build more factories?”
They are, but new capacity takes a year or more to build, qualify and ramp up. Most new supply arrives in 2027.

“Is the $90 to $425 figure cherry-picked?”
It’s one kit’s US retail price, so it’s illustrative. That’s why the report leans on contract prices from TrendForce, which show the same direction.

“So is AI a bubble or not?”
Neither side is clearly right. Even the “no bubble” analysis expects Big Tech’s returns to fall below target from 2028. Our view is a slowdown with write-downs, not a collapse, and we say that’s a judgement.

“Why should a normal person care?”
It already costs them: pricier PCs and phones, and RAM upgrades that cost four times as much.

“How reliable are the numbers?”
Most are forecasts or analyst estimates and may change. Some come via news sites quoting TrendForce or Gartner, whose full reports are paywalled. The report cites every source.

What’s still open

Review the report and script, and flag anything that doesn’t read clearly to you. Rehearse your part (Scene 6 to the conclusion, about 2.5 minutes), then record. Before using any number in public, check it against the report’s source list.